

The Dynamics of Mastery

Non-Linear Dynamics and Statistical Physics
of Human Skill Acquisition

Course: 194.068 Interdisciplinary Project in Data Science

Student: Aya Wahbi

Student ID: 01427598

Supervisor: Vito D.P. Servedio

Co-Supervisor: Florina Piroi

Domain-Specific Lecture: 166.142 Biology

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1. Abstract

This project investigates the statistical laws governing human skill acquisition using longitudinal competition data from the World Cube Association (WCA). Focusing on 3×3×3 Rubik's Cube solves, the study analyses 8.8 million individual timed attempts from a stratified cohort of 50 competitors spanning five skill tiers, each with over 200 official solves and careers exceeding two years.

Three interlocking research questions are addressed. First, individual learning curves are modelled by fitting power-law, exponential, and hybrid mixture models per solver using non-linear regression and BIC comparison. Clear model discrimination was achieved for 92% of the cohort (target: $\geq 60\%$). The power law emerged as the most parsimonious model overall (46% BIC-preferred), but a striking skill-level gradient was discovered: elite solvers require more complex curve shapes, mid-range solvers follow classical power-law learning, and slower solvers increasingly exhibit exponential convergence toward fixed performance ceilings.

Second, the evolution of solve-time distributions across career phases reveals that the ex-Gaussian distribution dominates across all phases (AIC-preferred in 96% of phase-fits). Distributional shifts were detected in 92% of solvers (target: $\geq 50\%$), with excess kurtosis (81.7% median change) and skewness (55.6%) shifting most—indicating that, for the majority of solvers, expertise eliminates the heavy right tail of slow-outlier solves.

Third, Detrended Fluctuation Analysis reveals a median scaling exponent $\alpha = 0.57$, significantly exceeding the white-noise baseline of 0.5 for 74% of solvers, demonstrating weak but robust long-range correlations in performance fluctuations. No significant early-versus-late career change in α was detected (Wilcoxon $p = 0.23$), suggesting the correlation structure is a stable individual characteristic.

The project combines non-linear regression, Bayesian model comparison, distribution fitting, bootstrap goodness-of-fit testing, and time-series analysis in a reproducible Python pipeline, version-controlled via Git, following the CRISP-DM lifecycle.

2. Introduction and Motivation

2.1 The Power Law of Practice

A foundational finding in cognitive science is that performance improves as a power function of practice—the "Power Law of Practice" (Newell & Rosenbloom, 1981). Early practice yields large gains, later practice yields smaller ones, and there is no hard ceiling.

Heathcote, Brown, and Mewhort (2000) challenged this consensus, arguing that the power law appears when performance is averaged across individuals, but that single individuals actually follow exponential curves. Under a power law, improvement is scale-free; under an exponential, each person converges toward a fixed limit. Distinguishing between these models requires

individual-level data dense enough to fit curves reliably—precisely the kind available in WCA competition records.

Beyond the learning curve itself, two further questions arise: does expertise reshape the full distribution of performance (not just the mean)? And after removing the smooth learning trend, do the residual fluctuations carry temporal structure—or are they random noise?

2.2 Research Questions

RQ1 — Learning-Curve Shape: How does the functional form of individual learning curves vary across solvers, and what solver characteristics are associated with different model structures?

RQ2 — Distributional Evolution: How does the statistical shape of solve-time distributions change across career phases, and which features shift most between early and late stages?

RQ3 — Fluctuation Structure: How does the temporal correlation structure of performance fluctuations differ between early-career and late-career phases, and what does this reveal about the regularity of skilled performance?

3. Methodology

3.1 CRISP-DM Process

The project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM):

- **Business Understanding:** Research questions from the cognitive science literature; KPIs defined a priori.
- **Data Understanding:** WCA database explored (14 tables, 6.65M result rows). 3×3×3 event identified as target.
- **Data Preparation:** Solve attempts extracted, merged with dates/rounds, filtered (DNF/DNS removed), converted centiseconds→seconds, ordered chronologically: 8,800,344 individual timed attempts.
- **Modelling:** Three pipelines: non-linear curve fitting + BIC (RQ1), distribution fitting + AIC + bootstrap KS (RQ2), DFA + bootstrap CIs + surrogate tests (RQ3).
- **Evaluation:** Pre-specified KPIs. Robustness checks: CV, outlier sensitivity, alternative splits, surrogates.
- **Deployment:** Git version control; CSV + figure exports; fully reproducible with public WCA data.

3.2 Data Source and Preparation

Data: WCA Results Database Export. Four tables loaded: results, result_attempts, competitions, round_types. Restricted to 3×3×3 (1,801,014 results). After merging, filtering, converting: 8,800,344 individual timed solves. Each solve assigned chronological solve number within solver's career.

3.3 Cohort Selection

Criterion	Threshold	Purpose
Min. competitions	> 15	Longitudinal coverage
Min. solves	> 200	Curve fitting
Min. career span	> 2 years	Learning trajectory

Table 3.1. Inclusion criteria.

7,356 solvers eligible (from 279,061). 50 selected via stratified sampling across 5 skill tiers (10/tier, most data-rich).

Tier	N	Last Avg (s)	Solves
T1 (fast)	10	5.9–9.2	2,838–3,933
T2	10	9.4–10.5	2,254–3,424
T3	10	10.6–11.9	2,416–3,760
T4	10	12.0–14.4	2,055–3,779
T5 (slow)	10	14.4–24.9	1,699–4,264

Table 3.2. Stratified sampling summary.

3.4 Experiment Design and Reproducibility

- All models and comparison criteria specified before data analysis.
- KPI thresholds defined a priori ($\geq 60\%$ RQ1, $\geq 50\%$ RQ2, CIs for RQ3).
- Cohort selection fixed before analysis.
- Code version-controlled via Git (separate scripts per RQ).
- Random seed = 42 for all stochastic components.
- WCA database publicly available for full replication.

3.5 Tools and Environment

Python: Pandas, NumPy, SciPy, Matplotlib, Seaborn, PyMC, ArviZ, Statsmodels. Standard desktop workstation.

4. Results

4.1 RQ1: Learning-Curve Modeling

4.1.1 Candidate Models and Fitting

Three models fit per solver:

1. Power law: $t(n) = a \cdot n^{-b} + c$

- Exponential: $t(n) = a \cdot e^{-b \cdot n} + c$
- Hybrid: $t(n) = w \cdot a \cdot n^{-b} + (1 - w) \cdot d \cdot e^{-e \cdot n} + c$

Non-linear least squares via scipy (TRF, 20K iterations). Asymptote $c \geq 2.0s$; $w \in [0,1]$. Multiple initial guesses per model.

Comparison: BIC (primary, Kass-Raftery 1995 interpretation), AICc (secondary), 2-of-3 consensus, effect size, parameter identifiability ($SE < 100\times$). Diagnostics: runs test, heteroscedasticity, profile-likelihood for w , 80/20 CV, outlier sensitivity, smoothed robustness, PyMC Bayesian validation (10 solvers).

4.1.2 Results

KPI Assessment

KPI	Target	Result	Status
Clear discrimination ($\Delta BIC > 6$)	$\geq 60\%$	46/50 (92%)	PASS
AICc-BIC agreement	—	45/50 (90%)	Strong

Table 4.1. RQI KPI assessment.

KPI Status: PASS

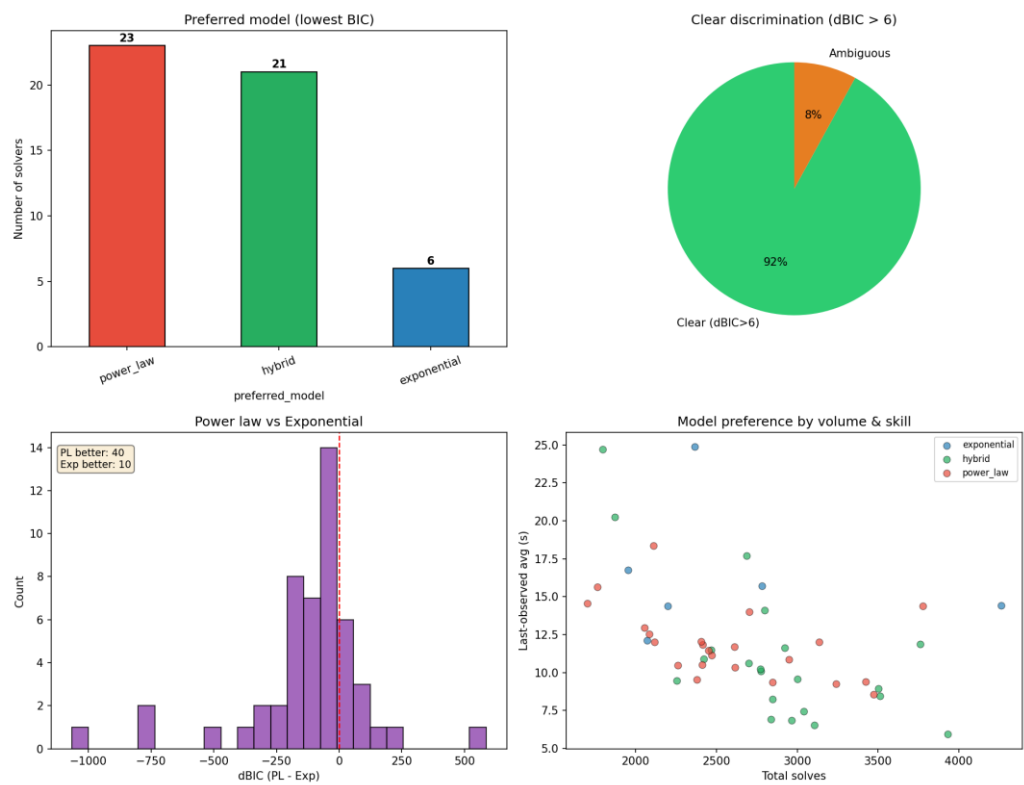


Figure 4.1. RQI summary dashboard

Model Preference

Method	Power Law	Hybrid	Exponential
BIC	23 (46%)	21 (42%)	6 (12%)
AICc	19 (38%)	26 (52%)	5 (10%)
Consensus	15 (30%)	30 (60%)	5 (10%)

Table 4.2. Model preference (N=50). Consensus biased toward hybrid (smoothed fits selected hybrid for 86%); BIC is the primary result.

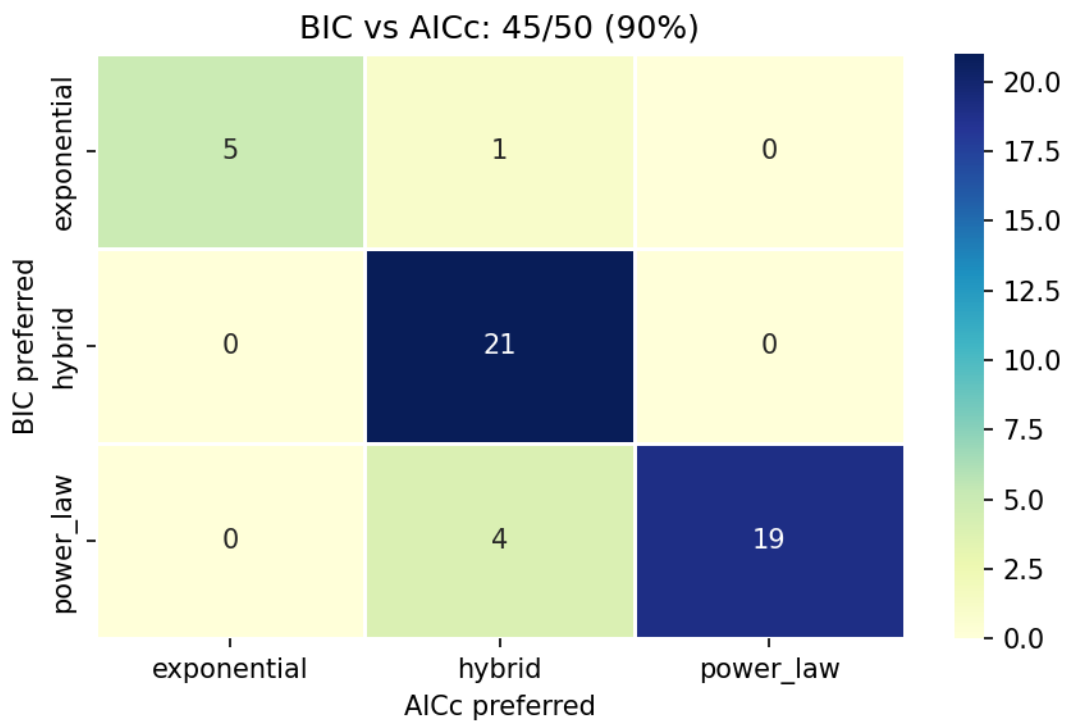


Figure 4.2. BIC vs AICc agreement

Model Preference by Skill Tier

Tier	Avg (s)	Hybrid	PL	Exp
T1	5.9–9.2	8	2	0
T2	9.4–10.5	4	6	0
T3	10.6–11.9	5	5	0
T4	12.0–14.4	1	7	2
T5	14.4–24.9	3	3	4

Table 4.3. BIC-preferred model by tier. Clear gradient: elite→hybrid, mid→PL, slow→exponential.

Goodness of Fit

Metric	Median	Mean	Range
R²	0.331	0.361	0.04–0.75
MAE (s)	1.41	1.56	0.65–3.84
Effect size	0.00%	1.12%	0–10.8%

Table 4.4. Fit statistics. Median effect size 0%: differences are statistically detectable but practically tiny.

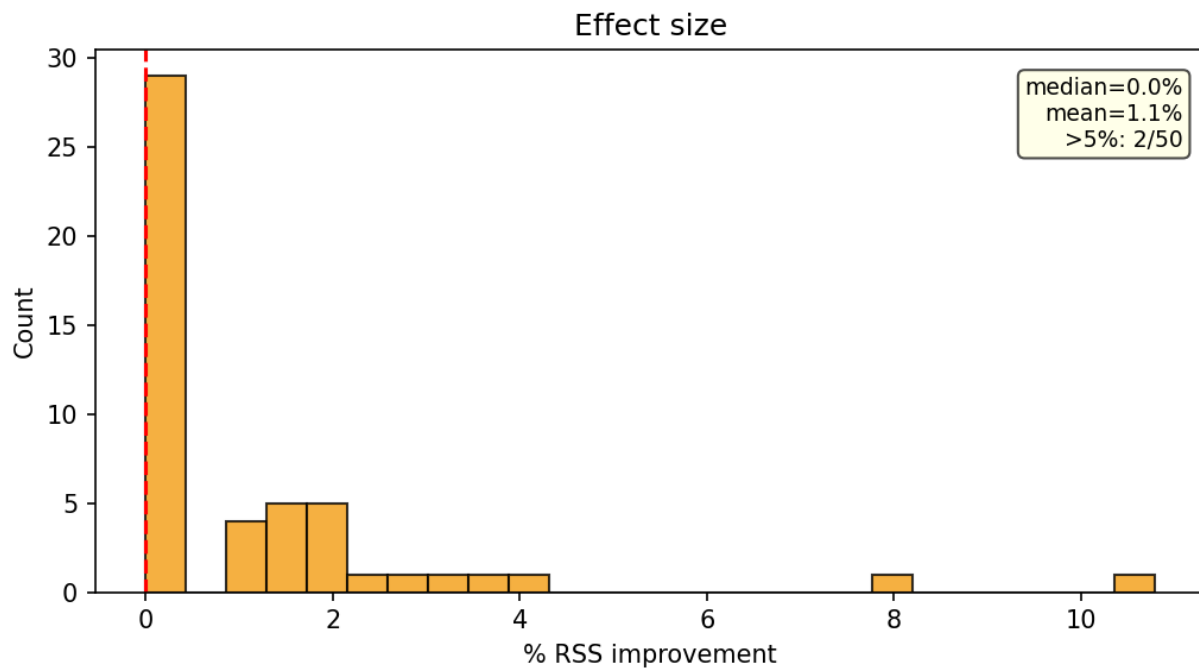


Figure 4.3. Effect size distribution.

Hybrid Identifiability

Model	Identifiable
Power law	50/50 (100%)
Exponential	50/50 (100%)
Hybrid	1/50 (2%)

Table 4.5. Parameter identifiability. Hybrid is overparameterised: median $w=0.82$, but w not constrained by data.

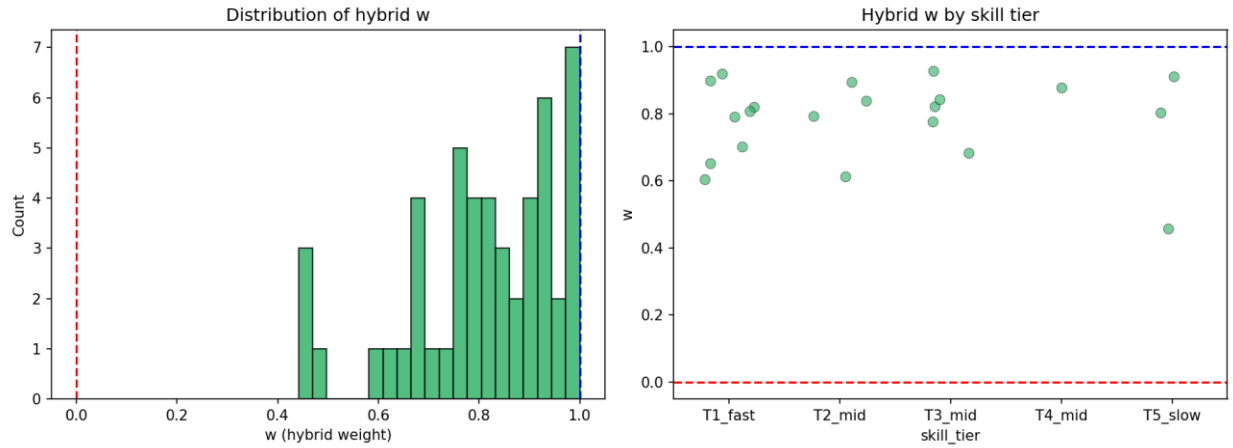


Figure 4.4. Hybrid w analysis

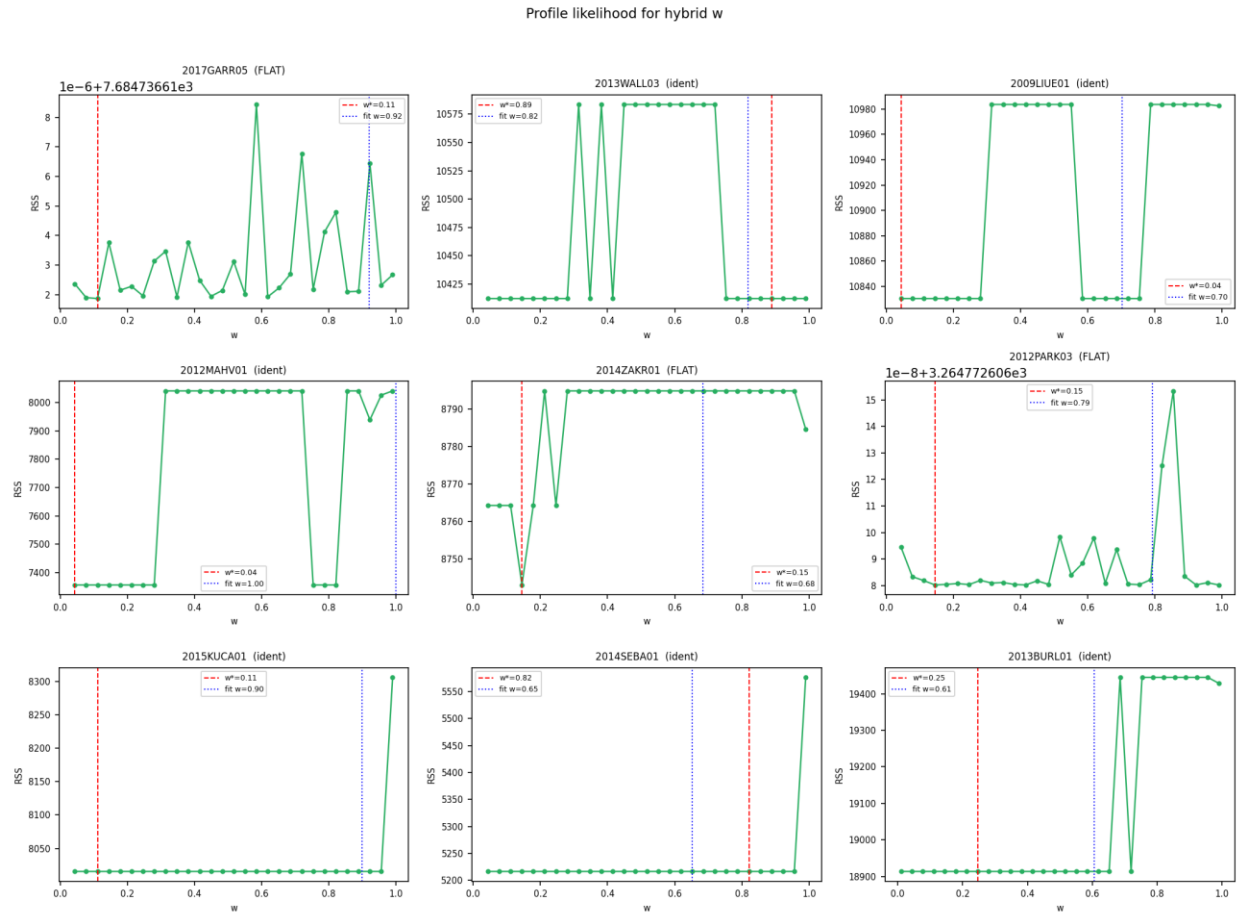


Figure 4.5. Profile-likelihood grids.

Residual Diagnostics

Diagnostic	Result
Runs test sig.	11/50 (22%)
Heteroscedasticity (decr.)	42/50 (84%)

No heteroscedasticity	8/50 (16%)
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Table 4.6. Residual diagnostics. Near-universal variance decrease: skill acquisition stabilises consistency.

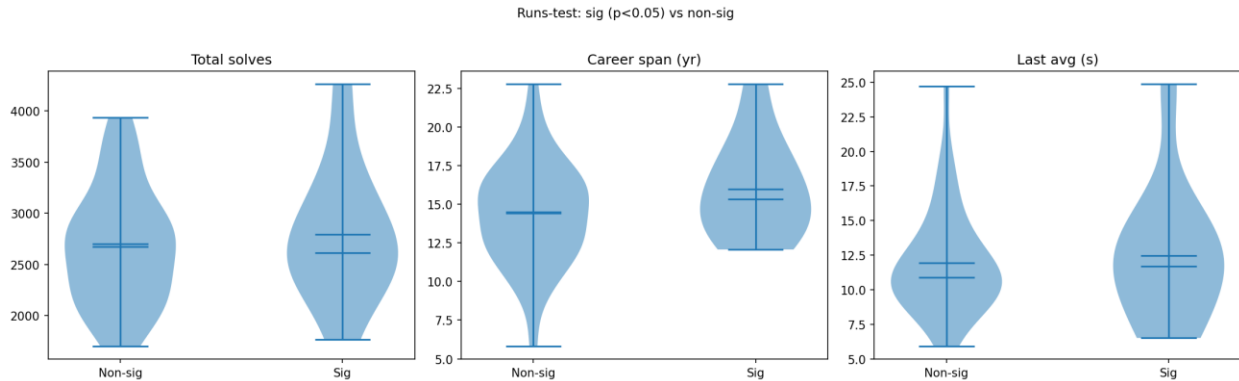


Figure 4.6. Runs test investigation

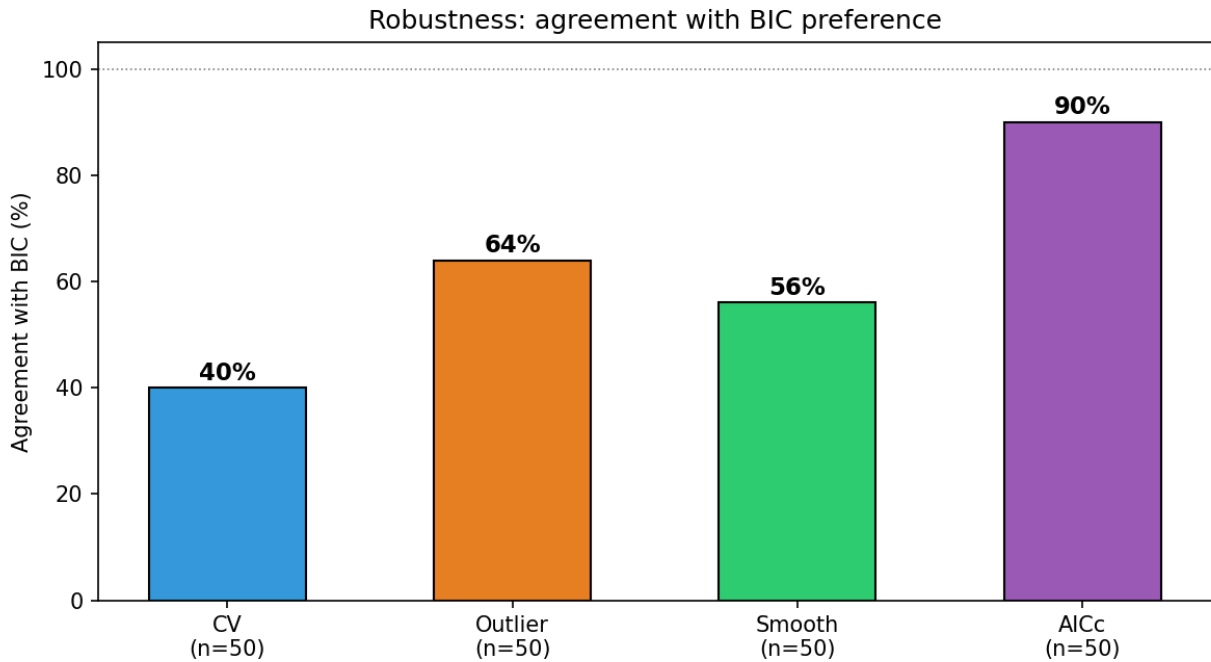


Figure 4.7. Robustness summary

Bayesian Validation

PyMC on 10 solvers: exponential sampled cleanly; power law had divergences (asymptote boundary); hybrid showed severe pathologies (2–423 divergences, ESS<100), confirming non-identifiability. Corroborates frequentist results but not standalone-valid.

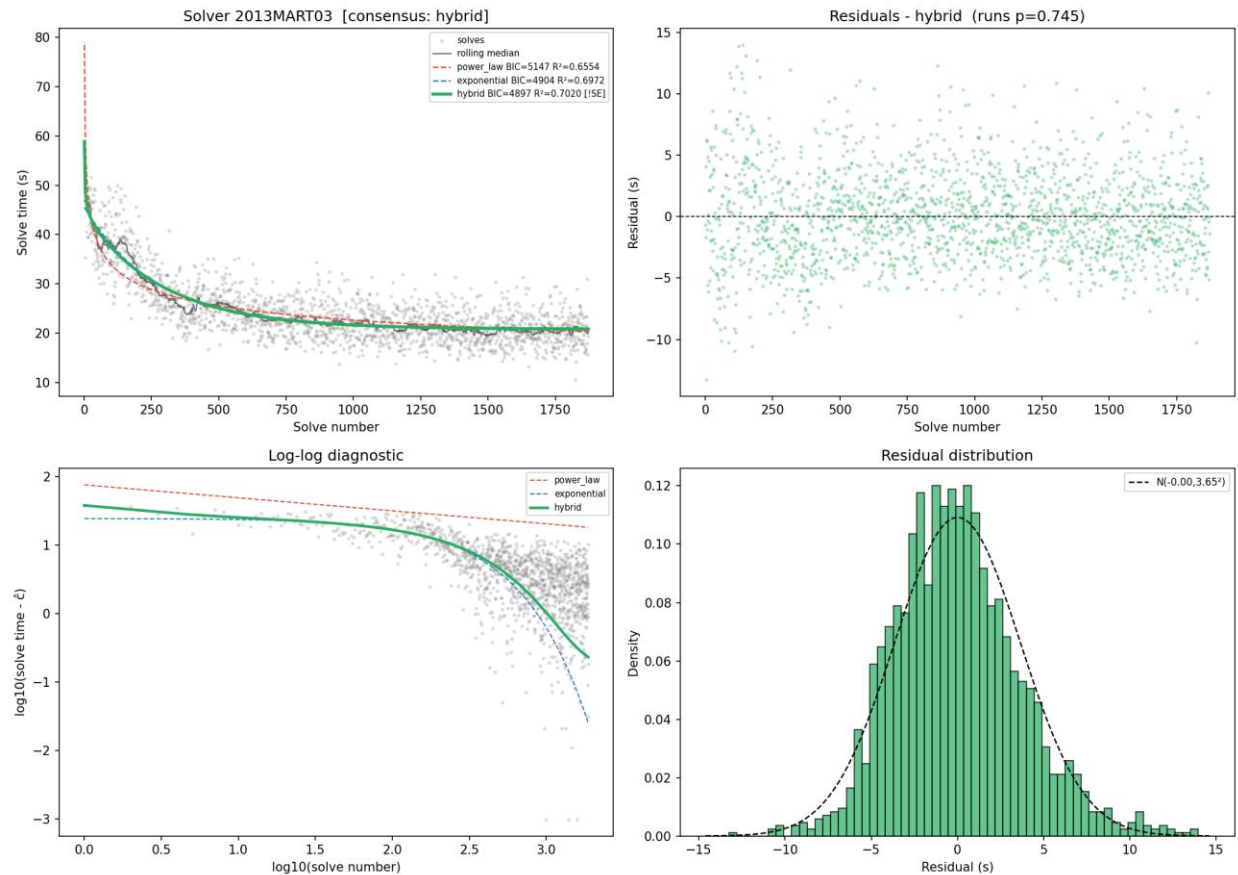


Figure 4.8–4.10. Example solver fits

4.1.3 Discussion

Neither the power law nor exponential is universally correct—the preferred form depends on skill level. Elite solvers need complex (hybrid) shapes; mid-range follow the power law; slower solvers converge exponentially. This reconciles Newell & Rosenbloom (1981) and Heathcote et al. (2000): both are correct for different populations.

The hybrid paradox: BIC-preferred for 42% but identifiable for only 2%. The appropriate interpretation is curve-shape complexity, not a specific mixture decomposition. With negligible effect sizes (median 0%) and universal non-identifiability, the power law remains the most parsimonious model.

Heteroscedasticity (84%) reveals two parallel processes in skill acquisition: improving mean and stabilising variance.

4.2 RQ2: Distributional Evolution

4.2.1 Overview and Methodology

Each solver's career split into quartiles. Three distributions fitted via MLE: Gaussian (2 params), log-normal (2 params), ex-Gaussian (3 params). Compared by AIC. Bootstrap KS goodness-of-fit (500 iterations). Shape vs scale shift separation via normalised KS test (z-scored Q1 vs Q4). Robustness: tercile, quintile splits.

4.2.2 Results

Ex-Gaussian Dominance

Phase	Gaussian	Log-Normal	Ex-Gaussian
Q1 (AIC)	0%	2%	98%
Q4 (AIC)	0%	6%	94%

Table 4.7. AIC-preferred distribution by phase.

Bootstrap KS: ex-Gaussian not rejected in 65.5% of phases vs 9.5% log-normal, 0.5% Gaussian.

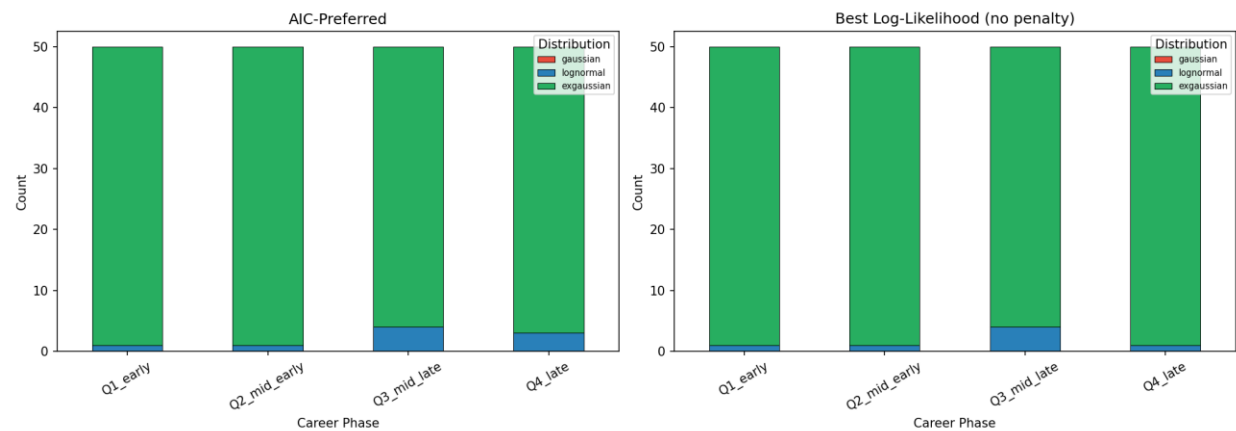


Figure 4.11. Distribution preference by phase

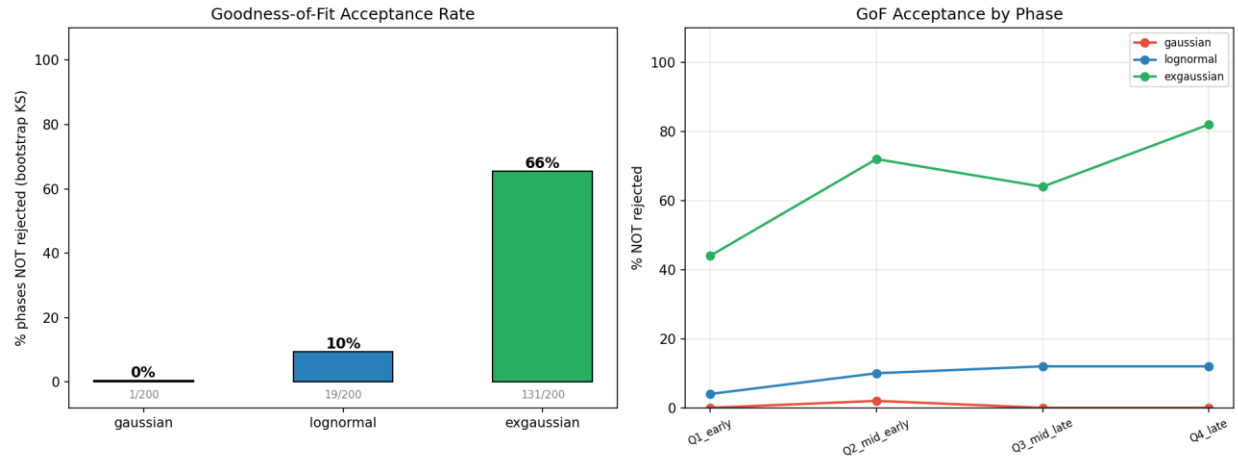


Figure 4.12. GoF acceptance rates

Shift Detection

Type	Count	%
Overall	46/50	92%
Family change	2/50	4%
Shape shift	46/50	92%
Scale shift	48/50	96%
Norm-KS reject	28/50	56%

Table 4.8. Shift detection summary.

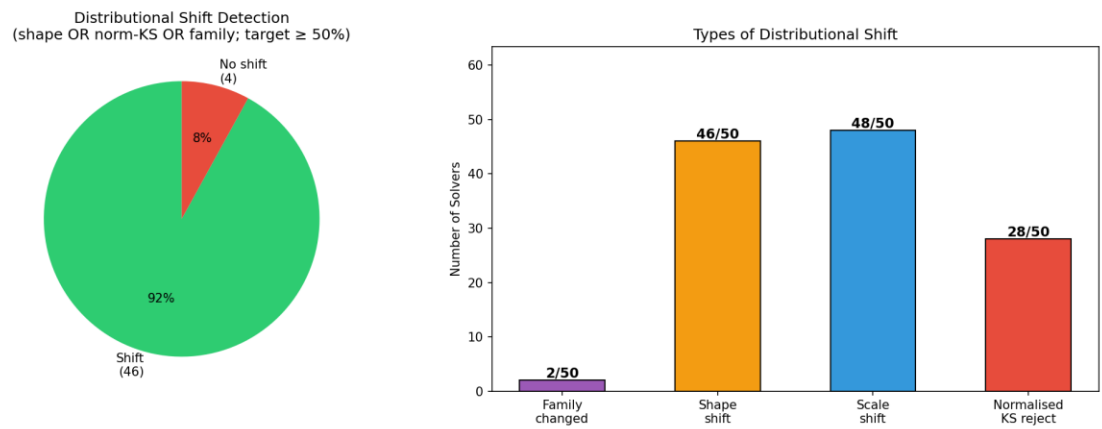


Figure 4.13. Shift detection

Feature Shift Ranking

Feature	Category	Rel. Change
Excess kurtosis	Shape	81.7%
Variance	Scale	76.1%

Skewness	Shape	55.6%
Std dev	Scale	51.2%
CV	Shape	36.2%
Mean	Scale	21.1%
IQR	Scale	37.7%
Tail weight (upper)	Shape	15.1%
P90/P10 ratio	Scale	10.9%

Table 4.9. Feature shift ranking Q1→Q4. Shape features (kurtosis, skewness) shift most.

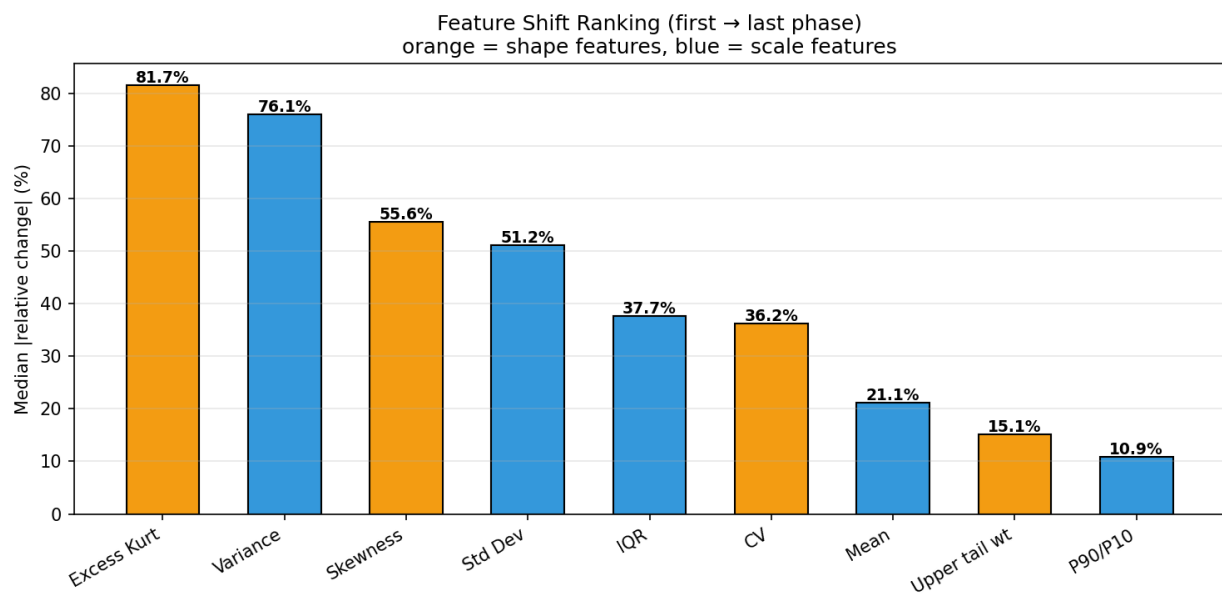


Figure 4.14. Feature shift ranking

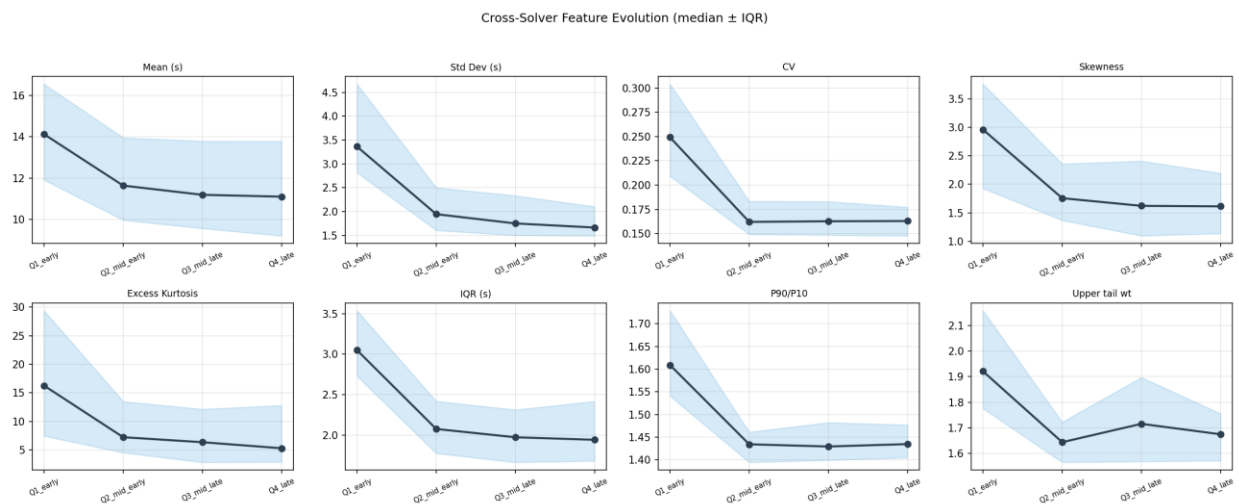


Figure 4.15. Feature evolution

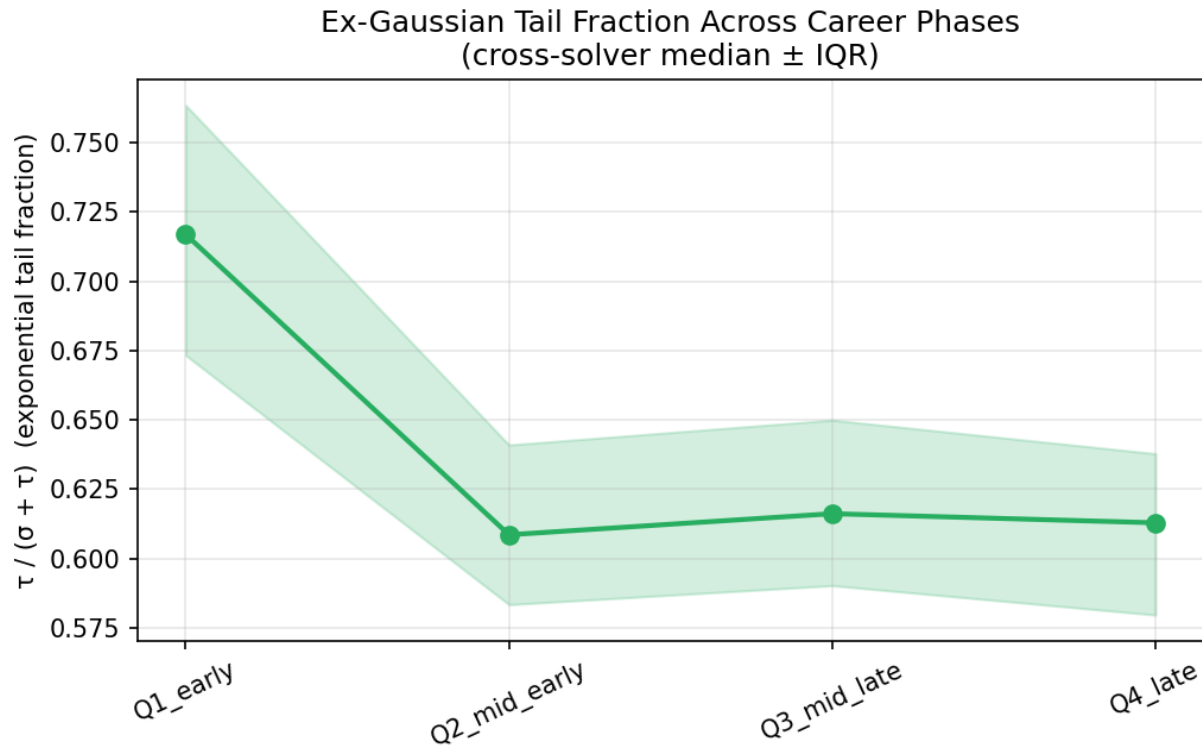


Figure 4.16. Tau fraction evolution

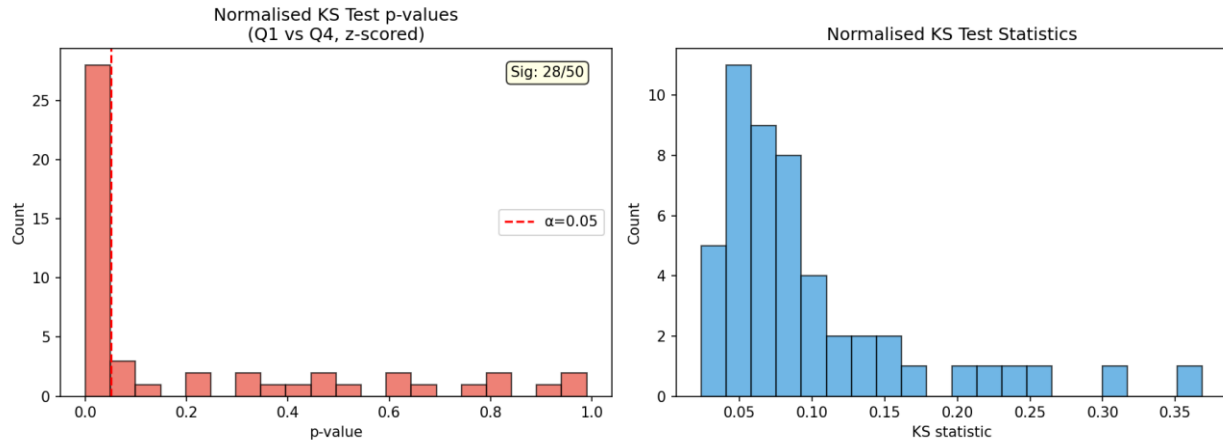


Figure 4.17. Normalised KS p-values

Robustness

Split	Shift	Agreement
Quartile	92%	—
Tercile	96%	96%
Quintile	98%	94%

Table 4.10. Robustness across splits.

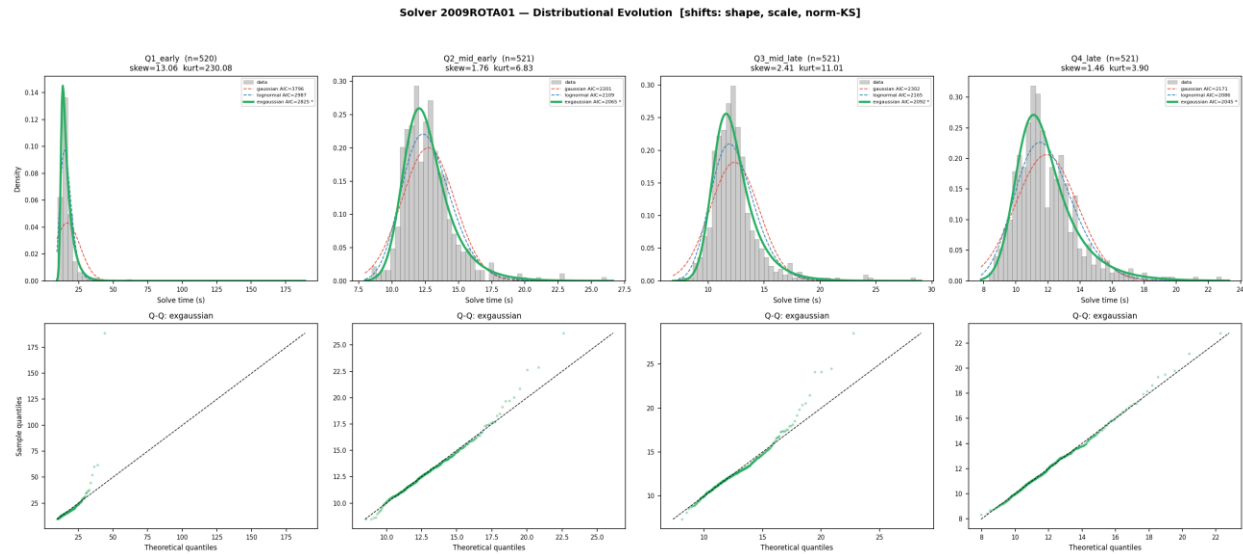


Figure 4.18–4.19. Example solver distributions and features

4.2.3 KPI Assessment

Target: $\geq 50\%$ shifts. Shape shift: 92%. Norm-KS: 56%. Both exceed target.

KPI Status: PASS

4.2.4 Discussion

The ex-Gaussian dominance extends reaction-time literature to speedcubing. For the majority of solvers, expertise eliminates the heavy right tail (kurtosis -82% , skewness -56%), not just compresses the distribution. Distributional family is stable (96% showed no change in distributional family)—parameters change, not form. $\sim 12\%$ showed increasing skewness, possibly reflecting plateaus.

4.3 RQ3: Fluctuation Structure (DFA)

4.3.1 Overview and Methodology

Residuals from RQ1 preferred model analysed via DFA (Peng et al., 1994). DFA-1 (linear detrending), windows 10 to 25% of series, 30 log-spaced points. Bootstrap CIs: 1,000 circular block-bootstrap replicates. Surrogate test: 200 random permutations. Career split at midpoint; Wilcoxon test on $\Delta\alpha$. PSD via Welch's method as consistency check.

4.3.2 Results

Full-Career Scaling Exponent

Statistic	Value
N	50
Median α	0.571
Mean α	0.579

Range	0.358–0.848
$\alpha > 0.5$	45/50 (90%)
Sig. (p<0.05)	37/50 (74%)

Table 4.11. Full-career DFA summary. Median $\alpha=0.57 > 0.5$ (white noise): weak but robust long-range correlations.

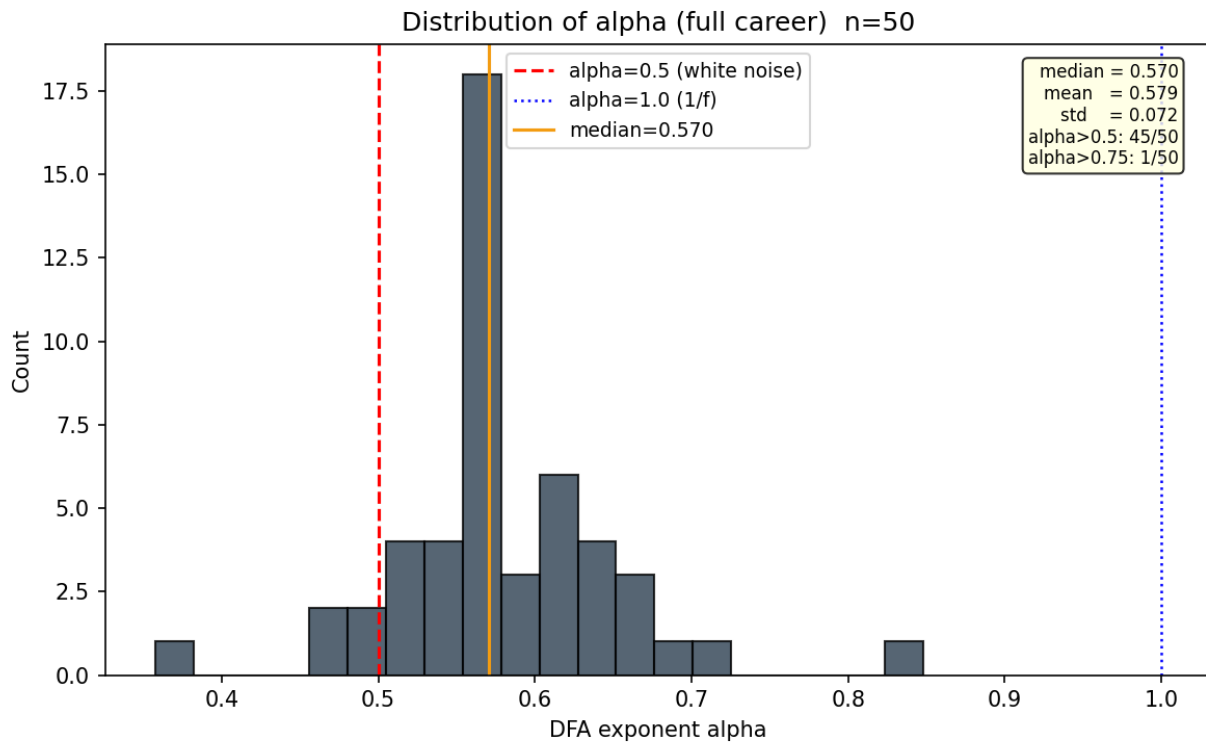


Figure 4.20. α distribution

PSD Consistency

Median $\beta = 0.027$; $\beta > 0$ for 76% (weak 1/f-like). Lower than theoretical $\beta=2\alpha-1 \approx 0.14$, likely due to Welch estimation in weak-correlation regime.

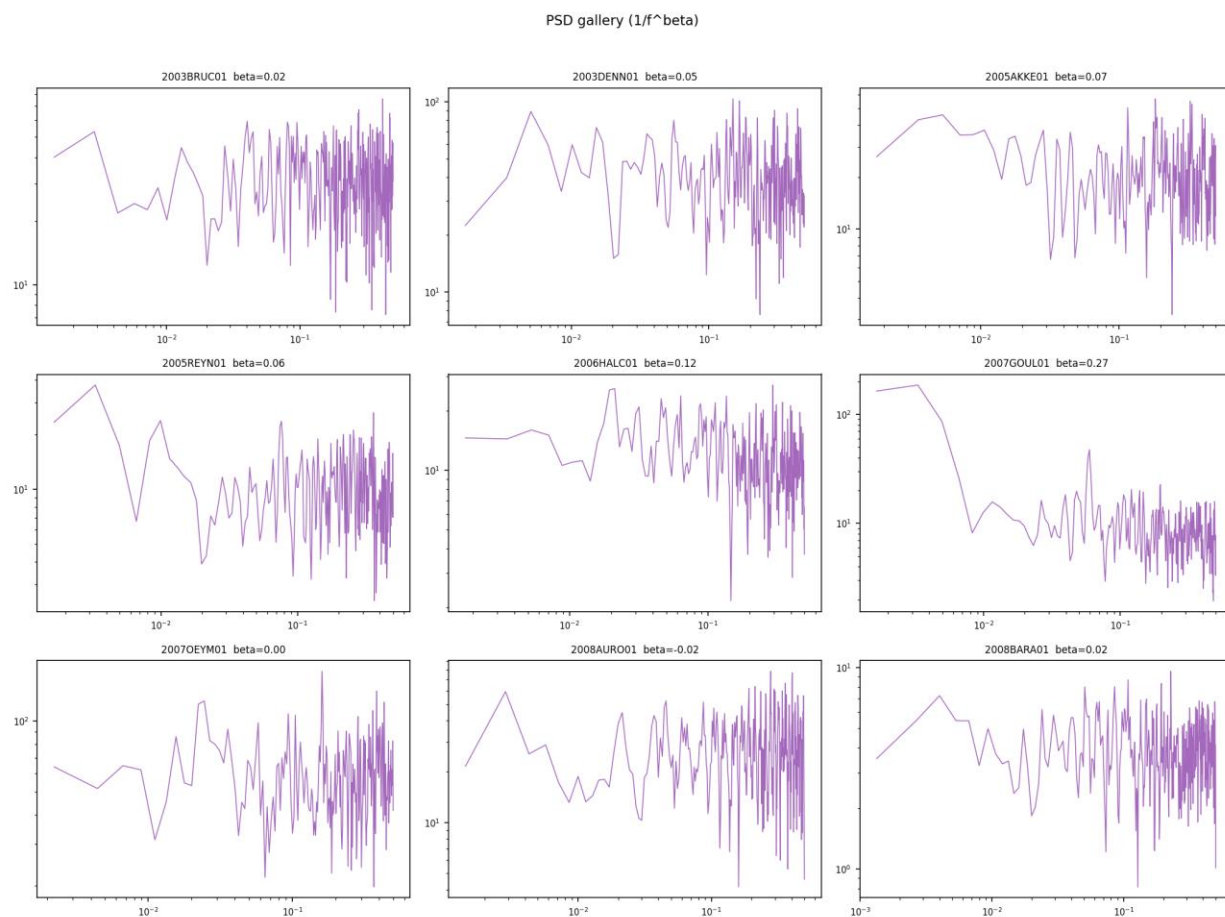


Figure 4.21. PSD gallery

Early vs Late Career

Statistic	Value
Median $\Delta\alpha$	-0.018
$\Delta\alpha > 0$	23/50 (46%)
Wilcoxon p	0.230

Table 4.12. No significant career-phase change. Correlation structure is stable.

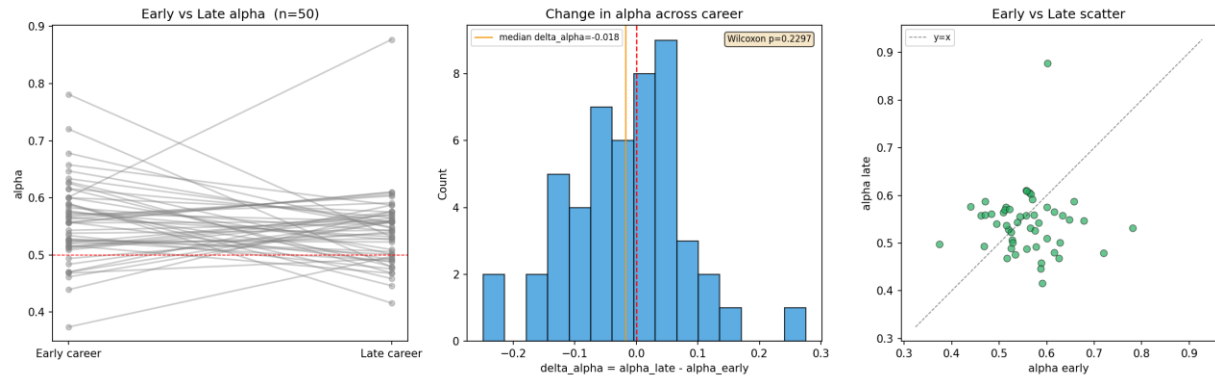


Figure 4.22. Early vs late career

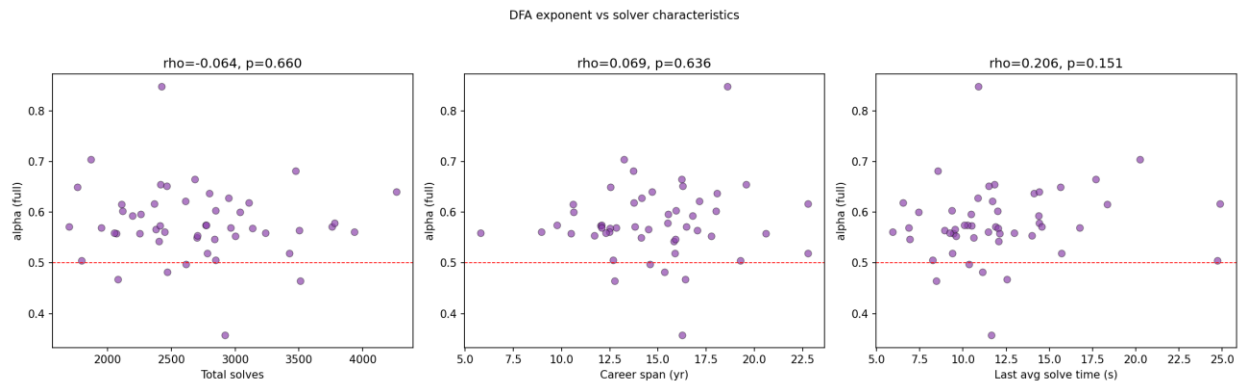


Figure 4.23. α vs characteristics

RQ3 - Fluctuation Analysis Summary

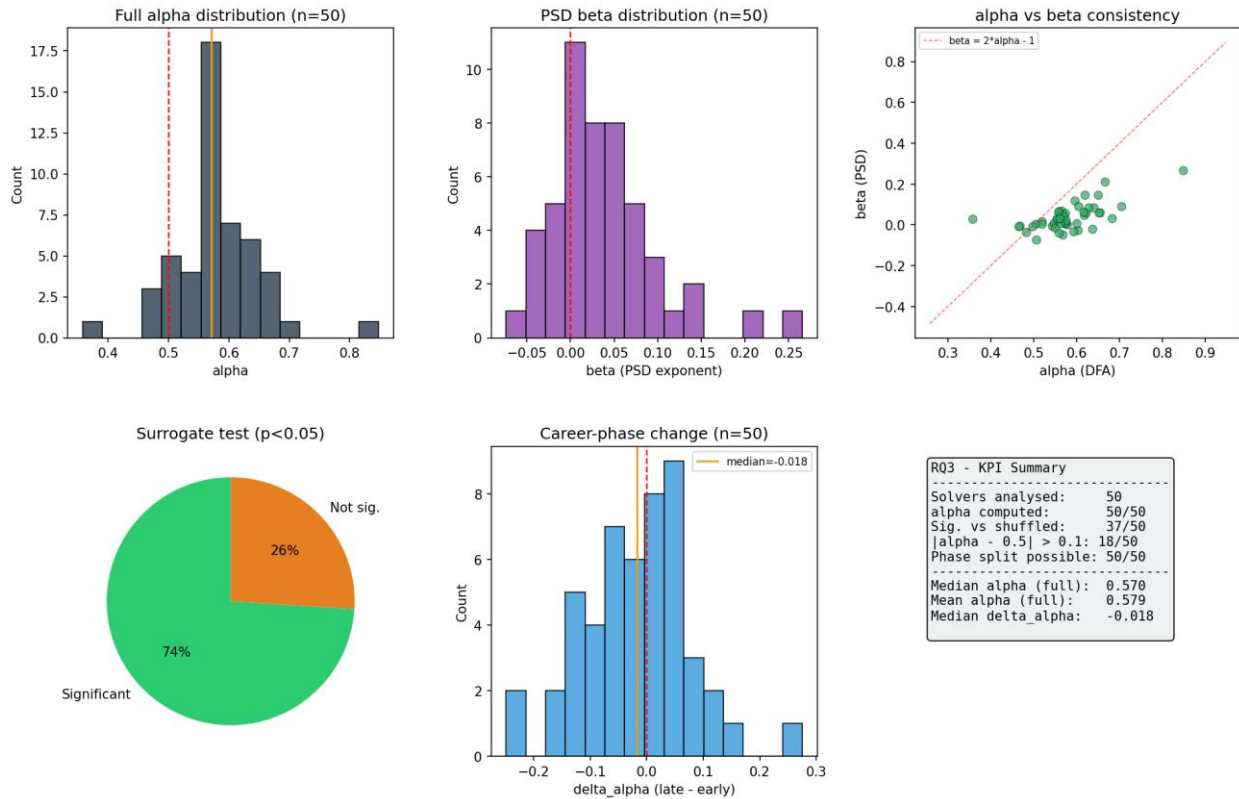


Figure 4.24. Summary dashboard out

4.3.3 KPI Assessment

- Exponent computed: 50/50 (100%)
- CIs available: 50/50 (100%, target $\geq 80\%$)
- Early/late comparison: 50/50
- Deviation from 0.5: median 0.571, sig. for 74%

KPI Status: PASS

4.3.4 Discussion

90% of solvers have $\alpha > 0.5$: performance fluctuations carry temporal memory ("streakiness"). Magnitude is modest ($\alpha = 0.57$, between white noise and strong 1/f). The stability across career ($p = 0.23$) suggests correlation structure is an individual trait, not an evolving skill feature—contrasting with RQ2's changing distributions.

Caveats: solve-order (not calendar) indexing; competition gaps unmodelled; residuals inherit RQ1 model misspecification; PSD β lower than theoretical prediction.

5. Cross-RQ Synthesis

The three RQs provide complementary views: RQ1 established the macro-level trajectory (skill-dependent functional form); RQ2 revealed micro-level distributional reshaping (tail elimination); RQ3 found stable temporal dynamics ($\alpha \approx 0.57$ throughout career).

Coherent picture: learning reshapes both central tendency (RQ1) and distributional form (RQ2), while temporal fluctuation dynamics remain a stable individual characteristic (RQ3). The heteroscedasticity from RQ1 (84%) is confirmed and deepened by RQ2: variance reduction involves genuine shape change. The 22% runs-test significance from RQ1 aligns with RQ3's temporal correlations.

6. Comparison to State of the Art

Learning-Curve Debate

Prior studies used limited laboratory data. This project analyses 1,700–4,200 points per solver. The novel finding—preferred form depends on skill level—offers a resolution: both power law and exponential are correct for different populations.

Distributional Analysis

Ex-Gaussian dominance extends reaction-time literature (Ratcliff 1979; Luce 1986; Matzke & Wagenmakers 2009) to speedcubing. Systematic τ decline with expertise parallels practice-effect findings.

Fluctuation Analysis

DFA applied to cognitive/motor performance adds a new domain alongside physiological signals (Peng et al. 1994), gait (Hausdorff et al. 1996), and cognition (Gilden 2001; Van Orden et al. 2003). The stability of α across career is a novel finding.

7. Domain-Specific Lecture

Selected: 166.142 Biology.

8. Conclusion

This project investigated skill acquisition using 8.8M solves from WCA. All three KPIs were met:

RQ1: 92% clear discrimination (target $\geq 60\%$). Functional form depends on skill level: power law (46% overall), but exponential for slow solvers and hybrid for elite. Power law is most parsimonious; hybrid is overparameterised (2% identifiable).

RQ2: 92% distributional shifts (target $\geq 50\%$). Ex-Gaussian dominates all phases. For the majority of solvers, expertise eliminates the heavy right tail (kurtosis -82% , skewness -56%). Distributional family is stable (96%).

RQ3: α computed with CIs for 100% (target $\geq 80\%$). Median $\alpha=0.571$, sig. for 74%. Stable across career ($p=0.23$).

Together: learning reshapes both mean trajectory and distributional form, while temporal fluctuation dynamics remain a stable individual characteristic.

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